

ES 446: Assignment 4

Reaction-Diffusion Equation Implementation in FEniCS

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1 Introduction

The objective of this assignment is to implement the Reaction-Diffusion equation on a unit square domain $\Omega = [0, 1] \times [0, 1]$. We utilize the FEniCS finite element computing platform to solve the resulting partial differential equation (PDE) over time.

2 Problem Statement

The governing strong form of the reaction-diffusion equation is given by:

$$\frac{\partial u}{\partial t} = D \nabla^2 u + f(u) \quad \text{in } \Omega \quad (1)$$

where:

- $u(x, y, t)$ is the concentration variable.
- D is the diffusion coefficient.
- $f(u) = \rho u(1 - u)$ is the reaction term (Fisher-KPP).

The boundary conditions are set as homogeneous Dirichlet:

$$u = 0 \quad \text{on } \partial\Omega \quad (2)$$

The initial condition is a Gaussian pulse centered in the domain:

$$u(x, y, 0) = \exp(-100 [(x - 0.5)^2 + (y - 0.5)^2]) \quad (3)$$

3 Derivation of the Weak Formulation

To solve the PDE using the Finite Element Method, we convert the strong form into its weak (variational) form.

3.1 Time Discretization

Using the Backward Euler method for the diffusion term and an explicit treatment for the reaction term (semi-implicit scheme):

$$\frac{u^{n+1} - u^n}{\Delta t} - D\nabla^2 u^{n+1} = f(u^n) \quad (4)$$

Rearranging for terms at time t^{n+1} :

$$u^{n+1} - \Delta t D \nabla^2 u^{n+1} = u^n + \Delta t f(u^n) \quad (5)$$

3.2 Variational Form

Multiply by a test function $v \in H_0^1(\Omega)$ and integrate over the domain Ω :

$$\int_{\Omega} u^{n+1} v \, dx - \Delta t D \int_{\Omega} (\nabla^2 u^{n+1}) v \, dx = \int_{\Omega} (u^n + \Delta t f(u^n)) v \, dx \quad (6)$$

Applying Green's first identity to the Laplacian term:

$$- \int_{\Omega} (\nabla^2 u^{n+1}) v \, dx = \int_{\Omega} \nabla u^{n+1} \cdot \nabla v \, dx - \int_{\partial\Omega} \frac{\partial u^{n+1}}{\partial n} v \, ds \quad (7)$$

Since $v = 0$ on the boundary $\partial\Omega$ (Dirichlet BC), the boundary integral vanishes. The final weak form is: Find $u^{n+1} \in V$ such that for all $v \in V$:

$$\int_{\Omega} (u^{n+1} v + \Delta t D \nabla u^{n+1} \cdot \nabla v) \, dx = \int_{\Omega} (u^n + \Delta t f(u^n)) v \, dx \quad (8)$$

4 Implementation

The following Python code using FEniCS implements the derived variational problem.

```
1 # Bilinear form a(u, v) and linear form L(v)
2 a = (u * v + dt * D * dot(grad(u), grad(v))) * dx
3 L = (u_n + dt * rho * u_n * (1 - u_n)) * v * dx
4
5 # Time-stepping loop
6 while t < T:
7     solve(a == L, u_sol, bc)
8     u_n.assign(u_sol)
9     t += dt
```

Listing 1: FEniCS Implementation

5 Results

The simulation results are captured in the plots below, showing the evolution from the initial Gaussian pulse to the diffused state.

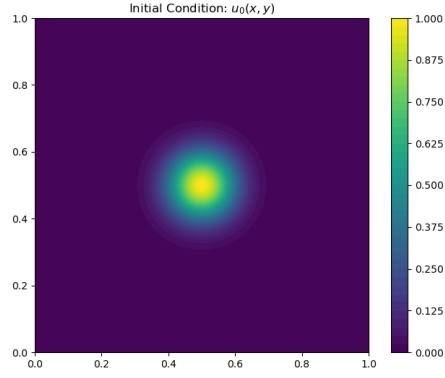


Figure 1: Initial Condition: Gaussian Pulse

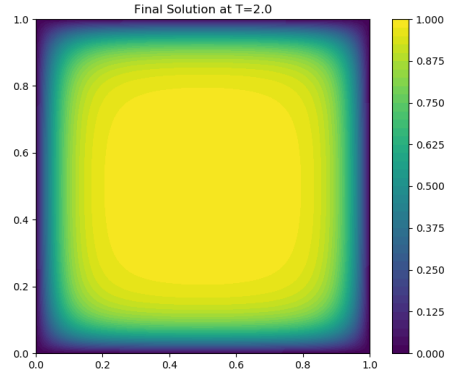


Figure 2: Final Solution at $T = 2.0$

6 Conclusion

The reaction-diffusion equation was successfully derived and implemented in FEniCS. The results demonstrate the physical behavior of concentration spreading while being constrained by zero-value boundary conditions.